

CONTACTS

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 Chicago, IL

SUMMARY

Senior Data Engineer with **8+ years** of experience designing and operating large-scale **ETL/ELT pipelines** across **logistics, e-commerce, healthcare and energy** and so on. **Expert in real-time and batch pipelines** using Spark, Flink, Airflow, and Beam across AWS, Azure, and GCP. **Specializes in building end-to-end data platforms**, including ingestion frameworks, transformation layers, dimensional models, and ML-ready datasets. **Strong collaborator** who works effectively with engineering, product, and business teams while mentoring others and driving measurable business impact.

EDUCATION

University of Missouri - Columbia

Master of Science in Computer Science

Columbia, MO 2015 - 2017

Missouri University of Science and Technology

Bachelor of Science in Computer Engineering

Rolla, MO 2011 - 2015

SKILLS

Programming & Scripting

Python · SQL · Scala · Java ·

Shell scripting · Go · Rust

Bobby Ray

Senior Data Engineer

EXPERIENCE

i3 Verticals

Senior Data Engineer

Nashville, TN

06/2023 - Present

- **Led real-time ETL/ELT logistics pipelines** using Apache Flink, Kafka, and Spark Streaming on Azure Event Hubs and Databricks, processing 3M+ daily telemetry and delivery events.
- **Build ingestion pipelines** using Kafka REST API and **automated relational DB ingestion into HDFS** via Sqoop, enabling scalable batch ETL workflows.
- **Developed streaming ETL flows** using **Kafka+Spark Streaming**, converting raw events into RDD/DataFrames and writing optimized Parquet datasets to HDFS.
- Designed and implemented enterprise **data warehouses** and **data marts** using **Kimball Methodology** (bus architecture, conformed dimensions, SCDs, surrogate keys, star/snowflake schemas).
- **Performed large-scale text analytics** and **ETL transformations** using Spark's in-memory engine with Scala.
- **Developed maintainable Python ETL code** for data processing, cleansing, enrichment, and analytics using modular libraries and reusable components.
- **Designed and deployed ETL/ELT workflows** using Airflow, PySpark and Azure Data Factory, transforming raw operational data into curated datasets for route optimization and driver performance.
- **Integrated predictive ML models** into streaming pipelines using Azure ML, DVC, and Azure Functions, enabling real-time ETA adjustments and reducing SLA breaches by 30%.
- **Provisioned ETL infrastructure** with **Terraform, Helm, and AKS**, automating multienvironment deployments and reducing setup time by 75%.
- **Built centralized observability for ETL pipelines** using ELK, Prometheus, Grafana, Azure Monitor, and Log Analytics, reducing MTTR by 50%.
- **Defined data contracts and KPIs** with product teams, and delivered self-serve analytics via Looker, Power BI, Synapse.
- **Integrated Power BI with SQL Server, Synapse**, and REST APIs to automate refresh cycles and ensure consistent ETL reporting.
- **Implemented Great Expectations** across 60+ Airflow DAGS, improving ETL data quality and reducing downstream issues.
- **Tuned Snowflake and BigQuery ELT workloads** using clustering keys, materialized views, and query optimization to reduce latency and cost.
- **Implemented CI/CD pipelines** for **dbt** models and Airflow DAGs, enabling automated testing and deployment of ETL/ELT logic.
- **Collaborated with product owners and analysts** in sprint planning to prioritize ETL platform features aligned with business KPIs.
- **Mentored junior engineers** in Agile best practices, including story slicing, estimation, and technical spike planning.

SKILLS

Data Engineering & ETL/ELT

ETL/ELT pipelines · batch & streaming ·
Spark · Flink · Beam · Airflow · dbt ·
Kafka · Kafka Connect

Data Warehousing & Databases

Snowflake · BigQuery · Redshift ·
Synapse · PostgreSQL · MySQL · Oracle ·
SQL Server · Cassandra · MongoDB ·
Hive · HBase · Redis · Elasticsearch

Cloud Platforms

AWS (Lambda, Glue, EMR, ECS, Kinesis) ·
Azure (Synapse, Databricks, Data
Factory) ·
GCP (BigQuery, Dataflow, Pub/Sub,
Vertex AI) ·
Kubernetes (AKS, EKS, GKE) · Terraform ·
Ansible

MLOps / DevOps

MLflow · DVC · Kubeflow · Jenkins ·
GitHub Actions · GitLab CI/CD · Docker ·
Helm · Prometheus · Grafana ·
CloudWatch · ELK Stack

Governance & Security

IAM · RBAC · GDPR · HIPAA · PCI-DSS ·
Encryption · Data Masking

Visualization & Analytics

Power BI · Tableau · Looker · Superset ·
Google Data Studio · Plotly · Dash

Collaboration & Version Control

Git · GitHub · GitLab · Bitbucket · Jira ·
Confluence · Agile/Scrum

EXPERIENCE

TurningPoint Healthcare Solutions

Mid-Level Data Engineer

Lake Mary, FL

09/2020 - 05/2023

- **Migrated 150+ legacy SQL Server ETL jobs to Apache Beam** on GCP Dataflow, improving performance 3x and reducing operational overhead.
- **Built real-time ETL pipelines for smart grid telemetry** using Flink, Kafka, and Pub/Sub, enabling fault detection across 10K+ nodes.
- **Developed dimensional models** using BigQuery, Cloud SQL, and dbt to support ELT transformations for predictive maintenance and outage tracking.
- **Led data governance initiatives** using GCP Data Catalog, Azure Purview, and schema registries to ensure compliant ETL/ELT data flows.
- **Created RDDs, DFs** and performed large-scale ETL transformations using PySpark.
- **Implemented CI/CD pipelines** with **GitHub Actions, Jenkins, and Cloud Build**, reducing deployment errors by 45% for ETL code.
- **Built KPI dashboards in Looker, Power BI, and Data Studio**, eliminating 50+ hours/month of manual reporting through automated ELT pipelines.
- **Delivered end-to-end data lineage and audit trails** using Unity Catalog, dbt, and GCP Logging, improving traceability of ETL/ELT processes.
- **Used agile metrics (velocity, cycle time)** to identify bottlenecks and improve throughput of ETL engineering tasks.
- **Drove adoption of data quality checks** and observability metrics as part of Definition of Done, increasing trust in ETL outputs.

Omnicell

Data Engineer

Mountain View, CA

10/2017 - 08/2020

- **Assisted and maintained ETL pipelines** using Python and SQL, supporting daily ingestion of customer and transaction data.
- Engineered **data lake ETL architecture** using **S3, Glue Catalog, and Athena**, to support analytics across marketing, sales, and fulfillment.
- **Optimized SQL queries and dashboards**, reducing query time by 30% and improving downstream ELT performance.
- **Developed RESTful APIs for real-time recommendations** using FastAPI, MongoDB, and Lambda, integrating them into broader ELT workflows.
- **Standardized validation workflows** using Great Expectations and PySpark on AWS Batch, improving ETL data quality scores by 30%.
- **Developed validation scripts to detect** missing values and schema mismatches across staging and production ETL pipelines.
- **Created ad-hoc reports** and extracts for business stakeholders using SQL and Excel.
- **Implemented CI/CD pipelines** with GitLab, Jenkins, and CodePipeline, reducing release cycles for ETL jobs from days to hours.
- **Tuned Snowflake and Redshift ELT workloads** using clustering keys and load testing with JMeter.
- **Built executive dashboards** in Tableau, Superset, and QuickSight, powered by automated ELT pipelines.
- **Shadowed senior engineers** during code reviews and sprint demos to learn best practices in ETL and data engineering.